

Crew Single Agent Formatted

CrewAI Single

"""

CrewAI Single Agent Example: Content Creator
A specialized agent for content creation tasks.

"""

```
from crewai import Agent, Task, Crew, Process
from langchain_openai import ChatOpenAI
from langchain.tools import Tool
```

```
# Initialize LLM
```

```
llm = ChatOpenAI(
    model="gpt-4",
    temperature=0.7,
    api_key="your-api-key-here"
)
```

```
# Define custom tools
```

```
def web_search_tool(query: str) -> str:
```

```
    """Simulated web search"""
```

```
    return f"Search results for '{query}': [Relevant information about {query}]"
```

```
def content_analyzer(text: str) -> str:
```

```
    """Analyzes content for readability and SEO"""
```

```
    word_count = len(text.split())
```

```
    return f"""Content Analysis:
```

```
- Word Count: {word_count}
- Readability: Good
- SEO Score: 85/100
- Suggestions: Add more keywords, improve meta description
"""
```

```
# Create tools
```

```
search_tool = Tool(
    name="WebSearch",
    func=web_search_tool,
    description="Search the web for information on any topic"
)
```

```
analyzer_tool = Tool(
    name="ContentAnalyzer",
    func=content_analyzer,
    description="Analyzes content for quality, readability, and SEO optimization"
)
```

```
# Create a single agent
```

```
content_creator = Agent(
    role="Content Creator",
    goal="Create engaging, informative, and SEO-optimized content",
    backstory="""You are an experienced content creator with expertise in:
    - Writing compelling articles and blog posts
    - SEO optimization
    - Audience engagement
    - Content strategy
```

```

    You have a talent for explaining complex topics simply and creating
    content that resonates with readers while achieving business goals.""",
    tools=[search_tool, analyzer_tool],
    llm=llm,
    verbose=True,
    allow_delegation=False # Single agent doesn't delegate
)

# Example 1: Blog post creation
def create_blog_post():
    print("\n" + "="*60)
    print("Example 1: Blog Post Creation")
    print("="*60 + "\n")

    task = Task(
        description="""Create a comprehensive blog post about 'The Future of AI
Agents'.

Requirements:
1. Research the topic using available tools
2. Write an engaging introduction
3. Cover key trends and developments
4. Include practical examples
5. Write a compelling conclusion
6. Optimize for SEO
7. Target length: 800-1000 words

```

```
        Analyze the content after writing to ensure quality.""",
        agent=content_creator,
        expected_output="A complete, SEO-optimized blog post with analysis report"
    )
```

```
crew = Crew(
    agents=[content_creator],
    tasks=[task],
    process=Process.sequential,
    verbose=True
)
```

```
result = crew.kickoff()
```

```
print("\n" + "="*60)
print("BLOG POST RESULT:")
print("="*60)
print(result)
```

```
return result
```

```
# Example 2: Social media content
```

```
def create_social_media_content():
    print("\n" + "="*60)
    print("Example 2: Social Media Content")
    print("="*60 + "\n")
```

```
task = Task(
```

```
description="""Create a social media campaign for launching a new AI
product.
```

```
Create:
```

1. 3 Twitter/X posts (280 characters each)
2. 2 LinkedIn posts (longer form)
3. 5 relevant hashtags
4. 1 Instagram caption with emoji

```
Theme: Revolutionary AI agent platform for businesses
```

```
Tone: Professional but engaging
```

```
Focus: Benefits and innovation""",
```

```
agent=content_creator,
```

```
expected_output="Complete social media content package"
```

```
)
```

```
crew = Crew(  
    agents=[content_creator],  
    tasks=[task],  
    process=Process.sequential,  
    verbose=True
```

```
)
```

```
result = crew.kickoff()
```

```
print("\n" + "="*60)  
print("SOCIAL MEDIA CONTENT:")  
print("="*60)
```

```
print(result)
```

```
return result
```

```
# Example 3: Product description
```

```
def create_product_description():
```

```
    print("\n" + "="*60)
```

```
    print("Example 3: Product Description")
```

```
    print("="*60 + "\n")
```

```
task = Task(
```

```
    description="""Write a compelling product description for an AI-powered  
project management tool.
```

```
    Product features:
```

- Automated task assignment
- AI-driven deadline predictions
- Smart resource allocation
- Real-time collaboration
- Integrated analytics

```
    Create:
```

1. Catchy headline
2. 2-paragraph description
3. 5 key benefits (bullet points)
4. Call-to-action

```
    Analyze the content for marketing effectiveness.""",
```

```
    agent=content_creator,  
    expected_output="Complete product description with analysis"  
)
```

```
crew = Crew(  
    agents=[content_creator],  
    tasks=[task],  
    process=Process.sequential,  
    verbose=True  
)
```

```
result = crew.kickoff()
```

```
print("\n" + "="*60)  
print("PRODUCT DESCRIPTION:")  
print("="*60)  
print(result)
```

```
return result
```

```
# Example 4: Email campaign
```

```
def create_email_campaign():  
    print("\n" + "="*60)  
    print("Example 4: Email Campaign")  
    print("="*60 + "\n")
```

```
task = Task(  
    description=""Create a 3-email drip campaign for onboarding new users
```

to an AI agent platform.

Email 1 (Day 0): Welcome and getting started

Email 2 (Day 3): Key features and tips

Email 3 (Day 7): Success stories and next steps

Each email should include:

- Compelling subject line
- Personalized greeting
- Clear value proposition
- Strong call-to-action
- Professional signature

Keep emails concise and actionable."""

agent=content_creator,

expected_output="Three complete email templates"

)

```
crew = Crew(  
    agents=[content_creator],  
    tasks=[task],  
    process=Process.sequential,  
    verbose=True  
)
```

```
result = crew.kickoff()
```

```
print("\n" + "="*60)
```



```
print("EMAIL CAMPAIGN:")
print("="*60)
print(result)
```

```
return result
```

```
# Example 5: Multi-step content workflow
```

```
def multi_step_workflow():
    print("\n" + "="*60)
    print("Example 5: Multi-Step Content Workflow")
    print("="*60 + "\n")
```

```
# Task 1: Research
```

```
research_task = Task(
    description="""Research the topic: 'How AI Agents Transform Customer
Service'"""
```

```
    Use the web search tool to gather:
```

1. Current trends
2. Statistics and data
3. Real-world examples
4. Expert opinions

```
    Compile findings into a structured brief.""",
    agent=content_creator,
    expected_output="Comprehensive research brief"
)
```

```
# Task 2: Outline
outline_task = Task(
    description="""Based on the research, create a detailed outline for an
article.
```

```
    Include:
```

1. Title options (3 variations)
2. Introduction hook
3. Main sections with key points
4. Conclusion summary
5. SEO keywords

```
    Make it structured and logical.""",
    agent=content_creator,
    expected_output="Detailed article outline"
```

```
)
```

```
# Task 3: Write
```

```
writing_task = Task(
    description="""Write the complete article based on the outline.
```

```
    Requirements:
```

- Follow the outline structure
- 1000-1200 words
- Engaging and informative tone
- Include specific examples
- Natural keyword integration
- Strong intro and conclusion""",

```
        agent=content_creator,  
        expected_output="Complete article"  
    )  
  
# Task 4: Review and optimize  
review_task = Task(  
    description="""Review and optimize the article.  
  
    Use the ContentAnalyzer tool to:  
    1. Check readability  
    2. Verify SEO optimization  
    3. Ensure quality standards  
  
    Make final improvements based on analysis."""  
    agent=content_creator,  
    expected_output="Polished, optimized article with analysis report"  
)  
  
# Create crew with sequential tasks  
crew = Crew(  
    agents=[content_creator],  
    tasks=[research_task, outline_task, writing_task, review_task],  
    process=Process.sequential,  
    verbose=True  
)  
  
result = crew.kickoff()
```

```
print("\n" + "="*60)
print("FINAL OUTPUT:")
print("="*60)
print(result)
```

```
return result
```

```
# Run examples
```

```
if __name__ == "__main__":
    print("CrewAI Single Agent Examples - Content Creator")
    print("=" * 60)
```

```
# Uncomment the example you want to run:
```

```
create_blog_post()
# create_social_media_content()
# create_product_description()
# create_email_campaign()
# multi_step_workflow()
```

```
print("\n" + "="*60)
print("Content creation completed!")
print("="*60)
```